INCH-POUND
MIL-PRF-27/18E
25 September 2009
SUPERSEDING
MIL-PRF-27/18D
25 April 1988

PERFORMANCE SPECIFICATION SHEET

TRANSFORMER, POWER 5.76 VA, 60 HZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-27.

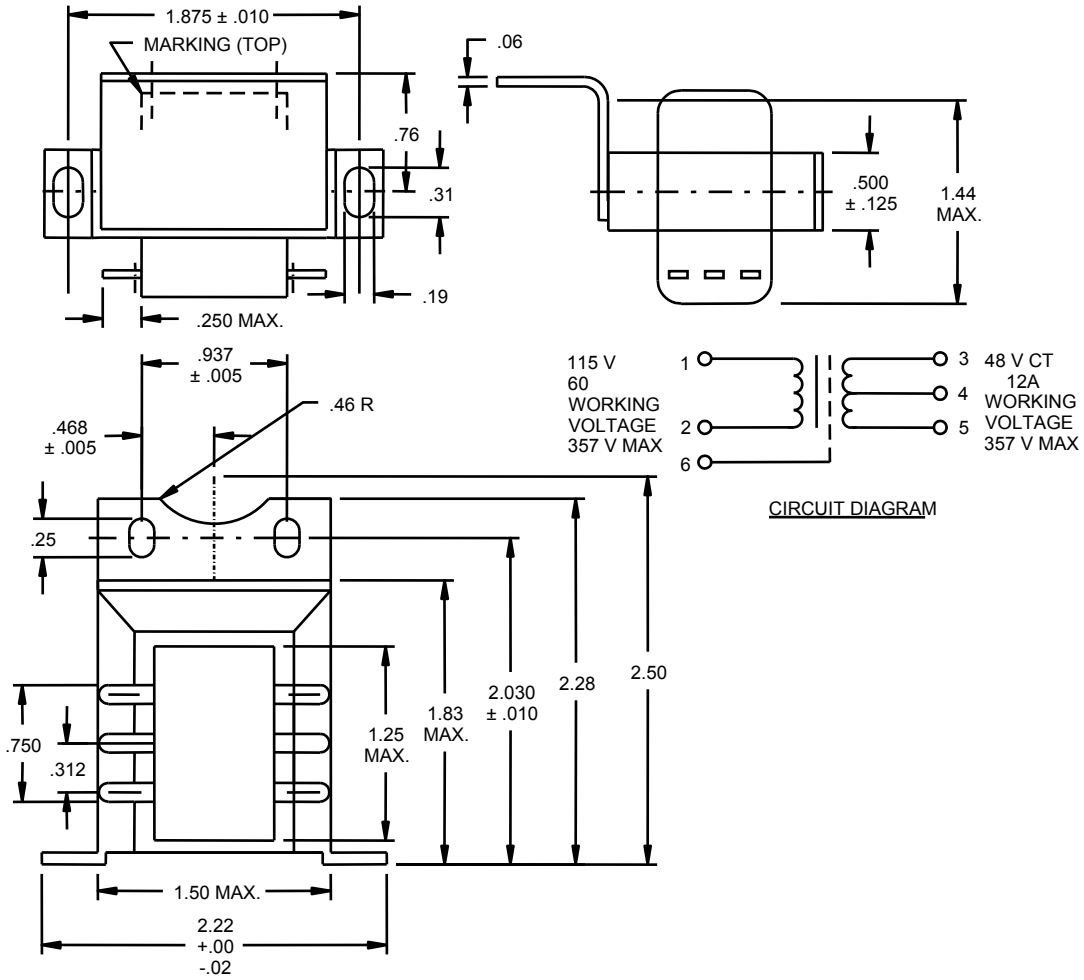


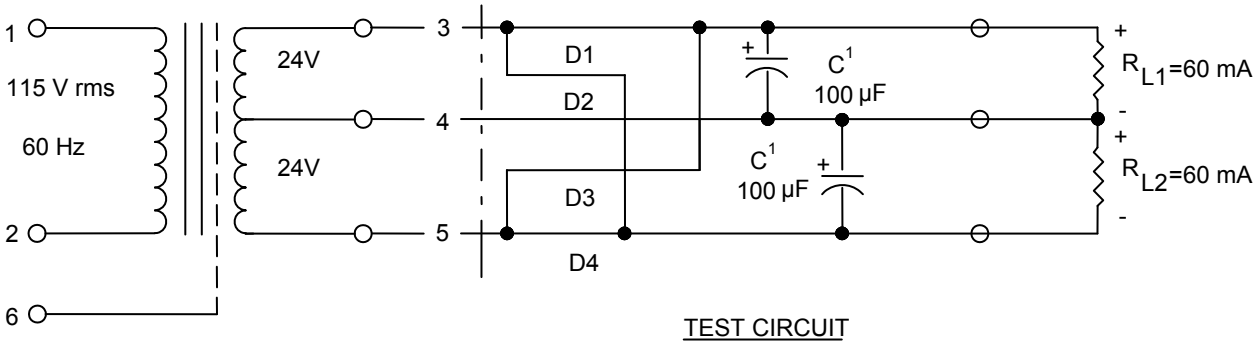
FIGURE 1. Dimensions and configuration.

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Inches	mm	Inches	mm
.005	0.13	.750	19.05
.010	0.25	.76	19.3
.02	0.5	.812	20.62
.06	1.5	.937	23.80
.125	3.18	1.25	31.8
.16	4.1	1.44	36.6
.19	4.8	1.50	38.1
.25	6.4	1.83	46.5
.31	7.9	1.875	47.63
.312	7.92	2.030	51.56
.46	11.7	2.22	56.4
.468	11.89	2.28	57.9
.500	12.70	2.50	63.5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.02 (0.5 mm) for two place decimals.
4. Tolerance on terminal position dimensions is ±.062 (1.57 mm).
5. Marking shall be on the top of the case.

FIGURE 1. Dimensions and configurations - Continued.



TEST CIRCUIT

WIRING DIAGRAM

NOTES.
1. D1, D2, D3, and D4 diodes are 1N5059.
2. C1 and C2 are 100 µF at 250 V dc.
3. R_{L1} and R_{L2} are 400Ω approximately. Adjust for 60 mA current.
4. Test circuit to be used when making rectified voltage measurements.

FIGURE 2. Test configuration.

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REQUIREMENTS: (When numbers in parentheses, such as (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-2): 5.8 voltamperes, 115 volts rms, 60 Hz.

Secondary voltage (3-5): 48 volts rms at rated load rectified.

Working voltage: (1-2) – 357 volts rms maximum.
(3-5) – 357 volts rms maximum.

Design, configuration, and physical dimensions: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: Solder type.

Height: .25 inch (6.35 mm) maximum.

Width: .125 ±.02 inch (3.18 ±0.51 mm).

Weight: .50 pound, maximum.

Operating temperature: -55°C to +130°C.

Turns:

Primary (1-2): 2,300 turns.

Secondary (3-5): 1,100 turns.

Ratio: 1:0.48.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 2 pounds.

Dielectric withstanding voltage (each winding) at sea level: MIL-STD-202, method 301; test voltage 1,000 volts rms.

Electrical characteristics:

No load: With 115 V rms and 60 Hz across (1-2), current in (1-2) shall not exceed 25 milliamperes, and the power in (1-2) shall not exceed 1.0 watts. Voltage across (3-5) shall be 55 volts rms ±2 percent.

Rated load: With 115 volts rms, 60 Hz across (1-2), and the rated current, the voltage across (3-5) shall be 48 volts rms ±3 percent, using rectified load shown on figure 2 (a .06 A dc, full wave bridge, capacitance filter).

DC resistance:

(1-2): 305 ohms maximum.
(3-5): 56 ohms maximum.

Efficiency: 70 percent minimum at rated load.

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Regulation: $\frac{\text{Voltage (no load)} - \text{Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$

Shall not exceed 21 percent.

Polarity - Additive, with terminals 2 and 3 connected.

Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Temperature rise: 40°C with 115 volts rms, 60 Hz across (1-2) at an ambient temperature of 90°C, full load terminals (3-5).

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/18-01.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air force – 85
DLA – CC

Preparing activity:
DLA - CC

(Project: 5950-2009-032)

Review activities:
Army – AR, CR4
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at http://assist.daps.dla.mil.

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